

LAJES AP, Portugal

WMO: 085090

Lat: 38.768N

Lon: 27.087W

Elev: 177

StdP: 14.60

Time Zone: -1.00 (AZR)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	48.0	49.8	39.2	35.5	52.5	41.0	38.1	53.3	40.6	59.1	34.7	57.9	6.1	320	0.442

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.7	81.0	72.1	79.2	71.2	78.2	70.6	73.6	78.5	72.6	77.3	71.8	76.3	11.8	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.9	118.6	76.1	71.2	115.9	75.6	70.0	111.2	74.7	37.1	78.6	36.3	77.6	35.5	76.4	79.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 31.1	26.8	23.4	DB	43.4	85.6	1.7	1.6	42.2	86.7	41.2	87.7	40.2	88.6	39.0	89.8	(4)
(5)			WB	41.9	75.9	1.9	1.4	40.5	76.9	39.4	77.7	38.4	78.5	37.0	79.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	64.2	58.6	58.2	58.8	60.2	62.9	66.9	71.0	72.9	71.1	66.9	62.5	59.8	(6)	
	DBStd	6.10	3.32	3.31	3.21	3.13	3.11	3.12	2.99	2.73	2.98	3.55	3.46	3.64	(7)	
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD65	1101	199	191	192	148	82	15	0	0	1	20	88	165	(9)	
	CDD50	5180	267	229	274	305	399	506	652	711	633	525	376	303	(10)	
	CDD65	806	0	0	1	3	15	72	187	246	184	81	14	3	(11)	
	CDH74	1962	0	0	0	1	5	76	477	899	429	73	2	0	(12)	
	CDH80	119	0	0	0	0	0	1	28	71	19	0	0	0	(13)	
(14) Wind	WSAvg	10.6	12.3	11.8	12.1	11.3	10.5	9.5	8.2	8.1	8.7	11.1	11.3	12.0	(14)	
(15) Precipitation	PrecAvg	45.4	6.3	5.4	5.2	3.0	1.8	1.9	1.2	1.9	2.6	4.3	4.5	7.2	(15)	
	PrecMax	59.8	14.1	15.8	11.1	5.4	5.6	4.1	2.7	4.7	6.0	9.8	10.9	16.6	(16)	
	PrecMin	26.3	1.5	0.4	0.7	0.3	0.5	0.7	0.1	0.2	0.7	1.3	0.6	0.6	(17)	
	PrecStd	8.9	3.4	4.1	2.8	1.8	1.3	1.0	0.8	1.1	1.5	2.6	2.8	4.7	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	68.0	68.2	69.7	71.4	74.6	78.7	82.6	84.4	82.3	78.4	73.6	69.7	(19)	
		MCWB	62.1	61.1	62.2	62.8	65.9	70.2	72.7	73.8	72.4	70.2	67.2	64.1	(20)	
	2%	DB	66.2	66.2	67.3	68.4	71.8	75.8	80.4	82.1	80.1	75.7	71.3	68.0	(21)	
		MCWB	61.6	60.5	60.5	61.7	64.5	68.3	71.4	72.7	71.7	69.0	65.6	63.4	(22)	
	5%	DB	64.7	64.6	65.5	66.6	70.0	74.0	78.5	80.2	78.4	74.0	69.6	66.5	(23)	
		MCWB	60.9	59.8	59.8	60.5	63.4	67.3	70.6	71.9	70.7	68.1	65.1	62.7	(24)	
	10%	DB	64.1	63.0	64.0	65.4	68.2	72.3	76.8	78.6	76.6	72.7	67.9	65.3	(25)	
		MCWB	60.5	59.0	59.2	59.8	62.2	66.1	69.6	71.1	69.7	67.4	64.0	61.8	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	63.9	63.1	63.7	65.0	67.4	71.5	74.3	75.6	74.3	71.9	68.7	65.7	(27)	
		MCDB	65.7	65.6	67.4	68.6	71.7	76.0	80.2	81.1	78.7	75.4	71.9	67.7	(28)	
	2%	WB	62.7	61.9	61.9	63.1	65.7	69.8	72.7	74.0	72.8	70.7	67.1	64.6	(29)	
		MCDB	65.0	64.7	65.3	66.6	70.0	74.2	77.6	79.2	77.3	74.1	69.8	66.7	(30)	
	5%	WB	61.8	60.8	60.8	62.0	64.6	68.6	71.7	73.0	71.9	69.5	65.9	63.5	(31)	
		MCDB	64.2	63.6	64.1	65.4	68.7	72.9	76.3	77.8	76.2	72.9	68.4	65.9	(32)	
	10%	WB	60.8	59.8	59.8	60.9	63.5	67.2	70.7	72.1	70.9	68.6	64.8	62.3	(33)	
		MCDB	63.4	62.6	63.1	64.2	67.5	71.3	75.2	76.9	75.0	71.9	67.4	64.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	8.1	9.0	8.9	9.3	9.7	10.0	10.3	10.7	10.0	9.2	8.0	8.1	(35)	
		MCDBR	8.7	10.1	10.5	11.2	11.6	11.8	12.4	12.9	12.2	10.8	9.8	8.8	(36)	
	5% WB	MCWBR	7.0	7.4	6.8	7.2	7.0	6.9	6.7	6.9	6.7	7.0	7.0	7.0	(37)	
		MCWBR	7.8	9.1	9.4	9.8	10.7	11.0	10.7	11.0	10.2	9.5	8.0	7.6	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.362	0.370	0.400	0.398	0.390	0.390	0.381	0.374	0.374	0.379	0.363	0.363	(40)	
		taud	2.456	2.439	2.347	2.379	2.439	2.473	2.509	2.551	2.552	2.507	2.485	2.428	(41)	
	Edn at Noon		255	270	272	279	283	282	283	283	278	263	253	244	(42)	
		Edh at Noon	26	30	36	37	36	35	33	31	30	28	25	25	(43)	
(44) All-Sky Solar Radiation	RadAvg		639	912	1274	1625	1905	2006	2042	1938	1537	1054	718	543	(44)	
	RadStd		49	49	90	135	128	140	112	73	79	55	38	36	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	+1.84	N/A	+0.63	+0.94	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	N/A	N/A	+1.03	N/A	+0.50	+0.77	N/A	N/A	N/A	N/A

Nomenclature: See separate page